

Richard B. Anderson

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EDUCATION

The University of Texas at El Paso, El Paso, TX

May 2026

Bachelor of Science in Physics, Magna Cum Laude

Bachelor of Science in Mathematics, Magna Cum Laude

- GPA: 3.81 / 4.00

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Fortran, Rust, Bash, Mathematica

Scientific Computing and Data: NumPy, SciPy, Pandas, Matplotlib, HDF5, TensorFlow, OpenCV

Modeling and Simulation: OpenMC, LIGGGHTS, Geant4, REBOUND, Simcenter STAR-CCM+, ParaView, MATLAB, Simulink

High-Performance Computing: Linux, Unix, SLURM, MPI, OpenMP, CUDA, parallel computing, computer hardware troubleshooting and maintenance

Engineering and Development Tools: Siemens NX, Teamcenter, ArduPilot, UAV hardware integration, CAD, Git, CMake, Make, GDB, Valgrind, Docker

TEACHING EXPERIENCE

Undergraduate Teaching Assistant, Electromagnetism Laboratory

El Paso, TX

The University of Texas at El Paso, Department of Physics

Aug 2025 – Present

- Guide students through electromagnetism lab setup, data collection, and post-lab data analysis.
- Troubleshoot experimental and conceptual problems while connecting measurements to course concepts.

Academic Tutor

El Paso, TX

El Paso Community College, Student Success

Jul 2022 – Aug 2023

- Tutored college mathematics from remedial algebra through differential equations in one-on-one and online settings.
- Adapted explanations to varied preparation levels, laying out step-by-step reasoning and problem setup.
- Maintained session records and coordinated with staff during high-demand periods.

TECHNICAL EXPERIENCE

Undergraduate Research Assistant

El Paso, TX

UTEP Astrophysics Theory Group | Advisor: Dr. Kedron Silsbee

Aug 2022 to May 2026

- Developed semi-analytic and numerical models of protostellar collapse and planetesimal capture using Python, C, and the REBOUND N-body simulation framework.
- Implemented custom numerical integration, interpolation, and ordinary-differential-equation solvers for analytically intractable physical models.
- Designed SLURM-based workflows for distributed simulation execution, automated data collection, and analysis of high-volume outputs on UTEP's high-performance computing cluster.
- Diagnosed numerical instability, verified simulation results, and built post-processing tools for orbital evolution, capture outcomes, and statistical analysis.
- Prepared a research manuscript and presented results at the APS Texas Section Fall Meeting in 2025.

SULI Intern, Nonproliferation and National Security Department

Upton, NY

Brookhaven National Laboratory

Jun 2025 to Aug 2025

- Developed an end-to-end computational workflow integrating LIGGGHTS discrete-element models, OpenMC radiation-transport simulations, CAD, Python data processing, and machine learning.
- Generated stochastic pebble-bed reactor geometries and incorporated them into Monte Carlo simulations to calculate neutron and gamma-ray spectra across fuel-burnup conditions.
- Built Python pipelines for simulation execution, data reduction, feature extraction, visualization, and preparation of machine-learning inputs.
- Trained TensorFlow models to estimate fuel burnup and reactor operating time from simulated gamma-ray spectra for nuclear-safeguards research.

Undergraduate Research Intern
UT Arlington | Neutrino and Rare Event Searches

Arlington, TX
May 2024 to Dec 2024

- Developed Python waveform-analysis pipelines for NEXT detector data to extract decay constants, peak widths, drift-time dependence, and pulse-shape observables.
- Applied SciPy signal-processing methods to denoise experimental waveforms, identify signal features, and compare reconstructed observables across operating conditions.
- Generated and analyzed alpha-particle events in Geant4/NEXUS to evaluate detector response, ionization transport, spatial distributions, and background signatures.
- Used the nuDoBE framework and Kolmogorov-Smirnov statistics to compare energy and angular distributions predicted by neutrinoless double-beta-decay models.

Undergraduate Research Assistant
UTEP Aerospace Center | Missile Systems Innovation Team

El Paso, TX
Sep 2023 to May 2024

- Supported development of a low-cost missile design pipeline using commercial off-the-shelf components, modular open-systems architecture, digital engineering, simulation, and rapid prototyping.
- Developed C++ and OpenCV training and deployment software for a computer-vision-based missile seeker, emphasizing modularity, computational efficiency, and repeatable testing.
- Built Python trajectory and aerodynamic-drag simulations to support missile sizing and launch-parameter optimization across subsonic, transonic, and supersonic regimes.
- Modeled thruster physics for simulation as part of an electron (ion) propulsion project at the Aerospace Center.
- Integrated trained computer-vision models with ArduPilot firmware, flight-controller hardware, and a UAV platform for onboard deployment, hardware testing, and system validation.
- Applied Siemens NX, Simcenter STAR-CCM+, and Teamcenter to CAD-based design review, computational analysis, and digital-engineering workflows.

PUBLICATIONS AND PRESENTATIONS

Richard B. Anderson and Kedron Silsbee: Capture of Interstellar Planetesimals. Manuscript in preparation

Richard B. Anderson and Kedron Silsbee: Capture of Freely Floating Planetesimals Into Protoplanary Disks. APS Texas Section Fall Meeting, 2025

HONORS AND AWARDS

Outstanding Senior in Physics: UTEP Department of Physics, 2026

NSF Graduate Research Fellowship Program Honorable Mention: National Science Foundation, 2026

Terry Foundation Scholarship: Terry Foundation, 2023 to 2027

NuPEERS Nuclear Physics Summer School: NuPEERS, 2023